Exam Date & Time: 02-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

Nanomedicine [BME 4306]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

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1a)	Given that a research scholar prepared three nanoparticle samples with diameters of 10 nm, 50 nm, and 70 nm, determine which sample would have the highest surface energy based on particle size. Explain your reasoning by applying principles of surface energy in nanoparticles	(2)
1b)	The 4th-century Roman glass cage cup, known as the Lycurgus Cup, exhibits different colors depending on the direction of light passing through it. Hypothesize the type of nanomaterials that might have been utilized in creating such glasses. Support your hypothesis with a comprehensive theoretical explanation.	(3)
1c)	Apply LaMer's theory to explain the process of nanoparticle formation, describing each stage and its significance in achieving controlled particle size.	(5)
2a)	Compare sonochemical and microwave synthesis methods for nanomaterial synthesis.	(2)
2b)	Summarize the sol-gel preparation method for nanomaterial synthesis and decide the drying condition to get aerogels.	(3)
2c)	A student aims to synthesize zinc oxide nanorods using the hydrothermal method, with zinc nitrate hexahydrate and sodium hydroxide as reactants. Develop the experimental setup and outline the procedure for this synthesis. Recommend key experimental conditions that should be adjusted to optimize the synthesis process.	(5)
3a)	Suggest the method to find out the crystallite size of a nanomaterial and explain the equation for the same.	(2)
3b)	A student prepared ZnO nanoparticles using Zinc nitrate and NaOH as reactants and functionalized it with oleic acid molecules. Suggest a characterization technique to understand the nature of this surface functionalization. Justify your suggestion with detailed explanation on the characterization technique.	(3)
3c)	Explain the Chemical Vapour Deposition method to synthesize nanostructures in detail	(5)

4a)	Suggest the characterization technique and procedure to obtain the excitation spectra of quantum dots.	(2)
4b)	Suggest a characterization technique to understand the hydrodynamic diameter of the nanoparticles. Explain the working principle of the instrument.	(3)
4c)	Develop a nanodrug delivery system in which pH triggers drug delivery.	(5)
5a)	Explain the significance of the first and second optical transparency windows for tissue in biomedical imaging?	(2)
5b)	Explain in detail the principle of photothermal therapy (PTT).	(3)
5c)	Design a nanoparticle contrast agent (a single nanosystem) for optical and MRI multimodal imaging. Justify the design.	(5)

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